Client Project Report: Income-Based Retail Marketing Targeting

1. Executive Summary

I recommend using the calibrated practical LightGBM model as the primary targeting model for this retail marketing use case. The model ranks individuals by estimated likelihood of annual income above $50K and converts the calibrated score into actionable campaign segments.

On the held-out test split, the recommended model achieved AP = 0.6953, ROC-AUC = 0.9550, F1 = 0.6341, Brier score = 0.0327, and ECE10 = 0.0024. The model is not presented as a perfect classifier. Instead, it is used as a practical ranking and segmentation model because it provides strong top-of-list concentration, calibrated scores, and actionable campaign tiers. For marketing activation, the most important result is the top-of-list performance: the top 5% segment reached 69.9% precision and 10.83x lift over the weighted base rate of approximately 6.5%.

The recommended business use is to start with the Top 5% premium segment for high-value outreach, expand to the Top 10% only after campaign-response measurement, and suppress the Bottom 50% from expensive outreach.

2. Business Problem and Data

The client wants to classify whether an individual has annual income above or below $50K and use the result for retail marketing. Because marketing budgets are limited, this is not only a binary classification problem. It is also a ranking and segmentation problem: the client needs to know which customers should be prioritized first, which customers should receive lower-cost outreach, and which customers should be excluded from high-cost campaigns.

The dataset contains census-style demographic, household, education, employment, and financial-signal variables. The target label is whether income is above $50K. The provided weight field is treated as a population/sample weight for training and evaluation, not as a predictive feature.

I frame the task this way because the client needs a model that supports an actual campaign decision, not just a classification score. The objective is not only to build a predictive model, but to connect model outputs to a business decision, a measurable operating policy, and a monitoring plan. In that context, I treat the score as a decision-support input for marketing prioritization, not as an automated eligibility, credit, employment, or other high-impact decision.

3. EDA and Preprocessing

The raw dataset contains 199,523 rows, 40 input columns, and a weighted population total of approximately 347.2M. The unweighted positive rate is 6.2% and the weighted positive rate is 6.4%, so the modeling problem is imbalanced. Accuracy alone would be misleading because a model could look strong by predicting the majority class. For this reason, the evaluation emphasizes PR-curve metrics, top-k precision, lift, and calibration.

Preprocessing decisions:

- Categorical values were cleaned, missing/unknown-style categories were explicitly handled, and rare categories were grouped to reduce overfitting.
- Employment, lifecycle, and financial-signal features were added, including weeks worked, full-year worker flag, employer-size grouping, investment-signal indicators, and log-transformed financial variables.
- Population weights were used as sample weights and were not used as predictive input features.
- Preprocessing artifacts were fitted only on training data and then applied to calibration/test data to avoid leakage.
- A separate calibration split was used for sigmoid calibration and F1-threshold selection.

4. Classification Model

Four model routes were retained because they serve different review purposes. Table 1 separates the final scoring model from the challenger and benchmark models, and Table 2 summarizes their held-out performance.

Table 1. Retained Model Routes and Client-Facing Roles

Model route	Role in this report	Why included	Client-facing use
Practical LightGBM	Recommended model	Best balance (ranking quality, calibration, and Top-K marketing lift)	Default model for campaign prioritization and score-tier segmentation
Monotonic LightGBM	Business-constrained challenger	To test whether directional business constraints can be added with limited performance loss	Use if the client prioritizes stronger directional business logic over maximum lift
Class-weighted logistic regression	Interpretable benchmark	Transparent linear benchmark with additional emphasis on the rare positive class	Use for coefficient-level review and sensitivity comparison
Population-weighted logistic regression	Transparent baseline	Simple weighted benchmark aligned with the provided sample/population weight	Use as the most transparent baseline for sanity checking

Table 2. Held-Out Test Performance of Retained Models

Model route	AP	PR-AUC	ROC-AUC	F1	Brier	ECE10	Top5 precision	Top5 lift
Practical LightGBM	0.6953	0.6952	0.9550	0.6341	0.0327	0.0024	0.6995	10.83x
Monotonic LightGBM	0.6869	0.6868	0.9537	0.6258	0.0339	0.0060	0.6864	10.63x
Class-weighted LR	0.6143	0.6145	0.9450	0.5792	0.0369	0.0044	0.6267	9.71x
Population-weighted LR	0.6237	0.6236	0.9458	0.5757	0.0365	0.0035	0.6362	9.85x

The final recommended model is Practical LightGBM (REFERENCE_practical_LightGBM) because it provides the strongest overall combination of ranking quality, probability quality, and top-of-list marketing lift. Its training recipe uses a LightGBM classifier with 500 estimators, learning rate 0.035, positive-class weighting with power 0.35, population weights, and sigmoid calibration on a separate calibration split. The most important practical model signals are exported in outputs/tables/practical_lgbm_feature_importance.csv and outputs/figures/ practical_lgbm_feature_importance_top20.png; the top drivers are weeks worked, investment signal, detailed occupation, age, education interactions, capital-income variables, and other demographic/employment variables. Because some demographic variables can raise privacy or fairness considerations, the client should confirm whether these variables are permitted for marketing use and may optionally run a sensitivity version excluding restricted demographic fields. The monotonic LightGBM model is retained as a business-constrained challenger. The logistic regression models are retained as transparent baselines for coefficient-level review.

5. Evaluation

The primary technical PR metric is Average Precision (AP). AP is reported because it summarizes precision-recall behavior across thresholds and is appropriate for imbalanced positive-class problems. I also added trapezoidal PR-AUC as a supplementary metric because some reviewers use the term PR-AUC directly. AP remains the primary PR-curve summary.

The main business metrics are Precision@TopK, Recall/CaptureRate@TopK, and Lift@TopK. These show how concentrated high-income individuals are in the ranked campaign list. For example, Precision@Top5 = 69.9% means that among the highest-scored 5% of individuals, roughly seven out of ten weighted individuals are true $50K+ cases. Lift@Top5 = 10.83x means this segment is more than ten times richer in $50K+ cases than the overall weighted population.

Probability quality is evaluated with Brier score, log loss, and ECE10. This matters because the client may use model scores not only for rank ordering but also for campaign thresholds, budget tiers, and monitoring. As shown in Figure 1, the practical LightGBM model provides the strongest lift across the main campaign cutoffs, especially around the Top 5% launch segment.

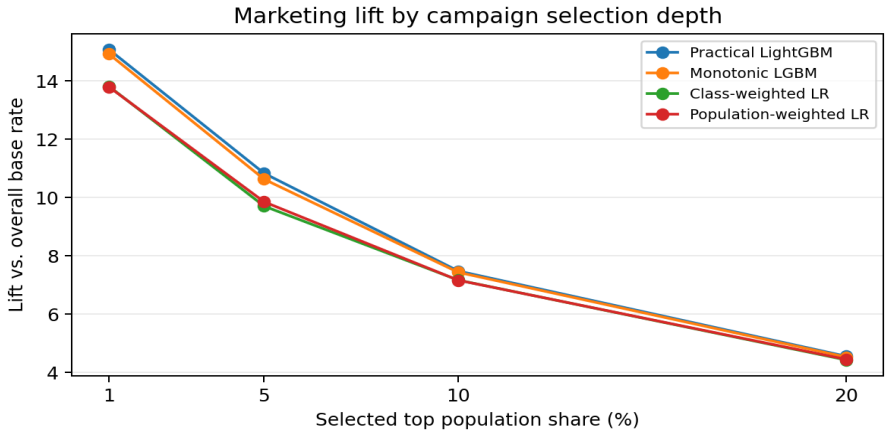


Figure 1. Marketing lift by campaign selection depth. The practical LightGBM model gives the strongest lift across the main campaign cutoffs, especially the Top 5% launch segment.

6. Segmentation

The main segmentation is a score-tier profile segmentation. It converts calibrated model scores into marketing tiers and then profiles each tier using demographic, employment, education, occupation, and financial-signal variables. This avoids treating segmentation as a simple score bucket. Each tier has a measurable profile and a recommended business action, summarized in Table 3.

Table 3. Score-Tier Marketing Segmentation and Recommended Actions

Segment	Population share	$50K+ rate	Lift	Recommended action
Top 5% premium	5.1%	69.9%	10.83x	Prioritize for high-value personalized outreach and premium offers.
Top 5-10% strong	5.1%	26.4%	4.09x	Use targeted offers with controlled acquisition cost and message testing.
Top 10-20% growth	10.0%	10.1%	1.56x	Use lower-cost digital campaigns and monitor incremental response.
Middle 20-50% broad	30.2%	1.5%	0.24x	Keep for broad awareness or small randomized tests, not expensive outreach.
Bottom 50% low-priority	49.5%	0.1%	0.01x	Suppress from high-cost campaigns; use only low-frequency broad messaging.

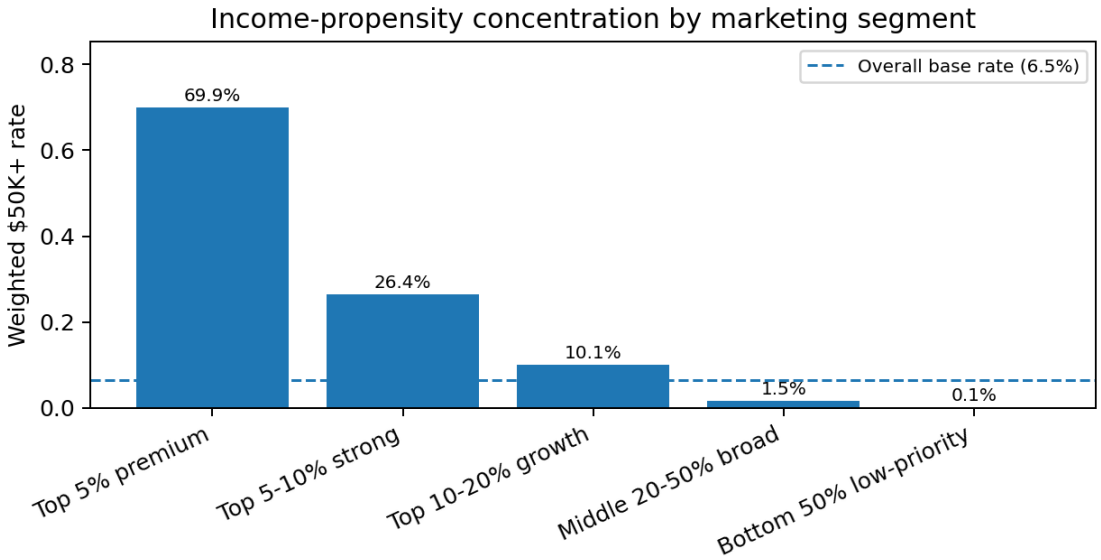


Figure 2. Weighted positive rate by marketing segment. The Top 5% segment is clearly separated from the rest of the population, supporting its use as the initial premium outreach group.

Figure 2 visually confirms that the Top 5% premium segment is clearly separated from the rest of the population by weighted $50K+ rate.

The Top 5% premium segment is the strongest activation group. It has the highest positive rate, the highest average score, strong full-year employment concentration, high college-plus share, high professional-occupation share, and strong investment-signal share. The Top 5-10% strong segment is still useful but should be handled with more cost discipline. The Top 10-20% growth segment is better suited to lower-cost digital campaigns and testing. The Bottom 50% low-priority segment should not receive expensive outreach.

An optional independent clustering segmentation is included in docs/segmentation_appendix.md. It excludes the target label and classifier score. The clustering diagnostics file now includes both inertia and silhouette diagnostics, but this clustering is not the main activation layer; it is a profile-discovery appendix to confirm that interpretable customer groups exist independently of the classifier score.

7. Recommendation

I recommend the following deployment strategy:

1. Use REFERENCE_practical_LightGBM as the default scoring model.
2. Launch the first campaign on the Top 5% premium segment.
3. Track response rate, conversion rate, revenue per contact, unsubscribe/complaint rate, and campaign ROI.
4. Expand to the Top 10% only if marginal ROI remains positive.
5. Use the monotonic model as a challenger if the client wants stronger directional business constraints.
6. Use the logistic models for transparent review and sensitivity checks, not as the primary scoring model.

Table 4 translates the model and segmentation results into an operating policy with success and guardrail metrics.

Table 4. Recommended Operating Policy and Success Metrics

Operating decision	Recommended approach	Success or guardrail metric
Initial launch scope	Start with Top 5% premium	Precision@Top5, response rate, revenue per contact, ROI
Expansion decision	Expand to Top 10% only after measuring incremental value	Marginal ROI, conversion lift, cost per conversion
Suppression decision	Exclude Bottom 50% from expensive outreach	Low positive rate, low expected campaign return
Monitoring	Review score distribution, segment mix, and calibration over time	ECE10, Brier score, segment share drift, campaign outcome drift
Responsible use	Use the score for marketing prioritization, not for eligibility, credit, employment, or other high-impact decisions	Use-case review, privacy review, complaint/unsubscribe guardrails

8. Limitations and Next Steps

This dataset predicts income, not actual retail purchase behavior. Income is a useful proxy for marketing potential, but it is not the same as customer lifetime value, purchase propensity, basket size, product preference, or campaign response. Because the dataset does not include retail transactions, campaign exposures, product affinities, or conversion outcomes, the segments should be interpreted as income-propensity marketing segments rather than complete retail customer personas. Before full deployment, the model should be validated against actual campaign outcomes.

Recommended next steps:

- Validate model lift on a real campaign-response dataset.
- Add customer-level transaction, browsing, and offer-response features if available.
- Monitor calibration drift and segment drift over time.
- Review privacy, compliance, and responsible-use requirements before activation.
- Run A/B tests comparing Top 5%, Top 10%, and broader control groups.

9. Implementation Readiness Checklist

Before activation, the client should confirm the operational items listed in Table 5. This keeps the report aligned with how the model would be reviewed and used in a business setting.

Table 5. Implementation Readiness Checklist

Area	Action owner	Readiness check
Business objective	Marketing/business owner	Campaign objective, budget, and success metric are defined before scoring.
Data pipeline	Data owner	Input schema, refresh cadence, and missing-value handling are documented.
Model use	Model owner	Score is approved for marketing prioritization only, not high-impact eligibility decisions.
Campaign test	Marketing operations	A/B or holdout design is prepared before expanding beyond the Top 5% segment.
Monitoring	Analytics/model owner	Score distribution, calibration, segment mix, response rate, conversion, and ROI are reviewed after launch.
Governance	Privacy/compliance reviewer	Privacy, opt-out, complaint, and responsible-use guardrails are reviewed before activation.

10. References Consulted

Business and marketing references

- McKinsey & Company. “What is personalization?” Used for business context on customer targeting, revenue lift, and marketing ROI.
- McKinsey & Company. “The value of getting personalization right or wrong is multiplying.” Used for personalization and revenue-impact context.
- Salesforce. “Data Driven Marketing.” Used for targeting, segmentation, campaign optimization, and ROI context.

Technical references

- scikit-learn documentation. average_precision_score, precision-recall evaluation, probability calibration, Brier score, common pitfalls/data leakage, KMeans, and silhouette analysis.
- LightGBM documentation. Booster model persistence and loading.
- NIST AI Risk Management Framework. Used as general guidance for risk management, transparency, monitoring, and responsible use.
- Mitchell et al. (2019). “Model Cards for Model Reporting.” Used as guidance for intended use, evaluation, and limitations.

Author’s related methodological background

- Bae, J., & Xi, Z. (2022). “Learning of physical health timestep using the LSTM network for remaining useful life estimation.” Reliability Engineering & System Safety, 226, 108717. Included only as related methodological background; the submitted model uses only the provided take-home data.